RESOLVING AT A PRIME WITH REGULAR LOCAL TARGET AND SEPARABLE RESIDUE EXTENSION

THE STACKS PROJECT AUTHORS

Editorial note. Inputs lemma-improve-presentation (07CE) and lemma-compare-standard (07EZ) are published entries used as standing inputs by their display numbers; not proven on the author's behalf.

Allowed standard prerequisites (editorial).

- Presentation improvement with retraction. lemma-improve-presentation (same chapter, PUBLISHED 07CE). Standardizes the auxiliary algebra B and freeness of the differential module.
- Comparison of standardness conditions. lemma-compare-standard (same chapter, PUBLISHED 07EZ). Promotes smoothness with free differentials to strict standardness of a power x_i^c.
- Parameter sequence with controlled annihilators. lemma-find-sequence (carded L26815). Supplies pi_1, ..., pi_d with pi_i^n in h_A and the annihilator equalities for e = 8c.
- Removing strictly standard elements from a resolution. lemma-lift-solution (carded L26812). Reduces the resolution to the quotient by the powers x_i^e.
- Height-zero delocalization. lemma-delocalize-height-zero (carded L26813). Reduces further to the localization at q.
- Ind-smoothness under nilpotent thickenings. proposition-lift (carded L26807). Identifies the flat Artinian quotient map as a filtered colimit of smooth algebras.
- Colimit of syntomic maps. Algebra, lemma colimit-syntomic (cross-chapter, named). Recognizes the flat map with separable residue extension as a filtered colimit of syntomic algebras.
- Colimit criterion for ind-smooth maps. Algebra, lemma when-colimit (cross-chapter, named). Extracts the solution from the colimit presentation.

Source reference. Original reference 07BV: focused-seeds/geometry/S9-smo-resolve-special/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07DY: focused-seeds/geometry/S9-smo-resolve-special/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:smoothing.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 07C3: focused-seeds/geometry/S9-smo-resolve-special/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07EZ: sources/stacks-smoothing/pinned-original-source.tex, source lines 936–936. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07FA: sources/stacks-smoothing/pinned-original-source.tex, source lines 2447–2447. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07FD: sources/stacks-smoothing/pinned-original-source.tex, source lines 2523–2523. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07CE: sources/stacks-smoothing/pinned-original-source.tex, source lines 559–559. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07F8: sources/stacks-smoothing/pinned-original-source.tex, source lines 2314–2314. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original proposition 07CM: sources/stacks-smoothing/pinned-original-source.tex, source lines 1591–1591. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07F7: sources/stacks-smoothing/pinned-original-source.tex, source lines 2296–2296. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $k \to A \to \Lambda \supset \mathfrak{q}$ be as in Situation 07F7 where*

(1) *k is a field,*
(2) *Λ is Noetherian,*
(3) *$\mathfrak{q}$ is minimal over $\mathfrak{h}_A$,*
(4) *$\Lambda_{\mathfrak{q}}$ is a regular local ring, and*

(5) *the field extension $\kappa(\mathfrak{q})/k$ is separable.*

Then $k \to A \to \Lambda \supset \mathfrak{q}$ can be resolved.

Proof. Set $d = \dim \Lambda_{\mathfrak{q}}$. Set $R = k[x_1, \ldots, x_d]$. Choose $n > 0$ such that $\mathfrak{q}^n \Lambda_{\mathfrak{q}} \subset \mathfrak{h}_A \Lambda_{\mathfrak{q}}$ which is possible as $\mathfrak{q}$ is minimal over $\mathfrak{h}_A$. Choose generators $a_1, \ldots, a_r$ of $H_{A/R}$. Set

$$B = A[x_1, \ldots, x_d, z_{ij}]/(x_i^n - \sum z_{ij} a_j)$$

Each B_{a_j} is smooth over R it is a polynomial algebra over $A_{a_j}[x_1, \ldots, x_d]$ and A_{a_j} is smooth over k. Hence B_{x_i} is smooth over R. Let $B \to C$ be the R-algebra map constructed in Lemma 07CE which comes with a R-algebra retraction $C \to B$. In particular a map $C \to \Lambda$ fitting into the diagram above. By construction C_{x_i} is a smooth R-algebra with $\Omega_{C_{x_i}/R}$ free. Hence we can find $c > 0$ such that x_i^c is strictly standard in C/R, see Lemma 07EZ. Now choose $\pi_1, \ldots, \pi_d \in \Lambda$ as in Lemma 07FD where $n = n$, $e = 8c$, $\mathfrak{q} = \mathfrak{q}$ and $I = \mathfrak{h}_A$. Write $\pi_i^n = \sum \lambda_{ij} a_j$ for some $\pi_{ij} \in \Lambda$. There is a map $B \to \Lambda$ given by $x_i \mapsto \pi_i$ and $z_{ij} \mapsto \lambda_{ij}$. Set $R = k[x_1, \ldots, x_d]$. Diagram

$$\begin{array}{ccccc} R & \longrightarrow & B & & \\ \uparrow & & \uparrow & \searrow & \\ k & \longrightarrow & A & \longrightarrow & \Lambda \end{array}$$

Now we apply Lemma 07F8 to $R \to C \to \Lambda \supset \mathfrak{q}$ and the sequence of elements $x_1^c, \ldots, x_d^c$ of R. Assumption (2) is clear. Assumption (1) holds for R by inspection and for Λ by our choice of $\pi_1, \ldots, \pi_d$. (Note that if $\mathrm{Ann}_\Lambda(\pi) = \mathrm{Ann}_\Lambda(\pi^2)$, then we have $\mathrm{Ann}_\Lambda(\pi) = \mathrm{Ann}_\Lambda(\pi^c)$ for all $c > 0$.) Thus it suffices to resolve

$$R/(x_1^e, \ldots, x_d^e) \to C/(x_1^e, \ldots, x_d^e) \to \Lambda/(\pi_1^e, \ldots, \pi_d^e) \supset \mathfrak{q}/(\pi_1^e, \ldots, \pi_d^e)$$

for $e = 8c$. By Lemma 07FA it suffices to resolve this after localizing at $\mathfrak{q}$. But since $x_1, \ldots, x_d$ map to a regular sequence in $\Lambda_{\mathfrak{q}}$ we see that $R_{\mathfrak{p}} \to \Lambda_{\mathfrak{q}}$ is flat, see Algebra, Lemma 07DY. Hence

$$R_{\mathfrak{p}}/(x_1^e, \ldots, x_d^e) \to \Lambda_{\mathfrak{q}}/(\pi_1^e, \ldots, \pi_d^e)$$

is a flat ring map of Artinian local rings. Moreover, this map induces a separable field extension on residue fields by assumption. Thus this map is a filtered colimit of smooth algebras by Algebra, Lemma 07BV and Proposition 07CM. Existence of the desired solution follows from Algebra, Lemma 07C3. □